

Project: Causal Machine Learning

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Question 1 - 2 - 4: Mark Vergouwen

Question 3 - 5: Milan Vervoort

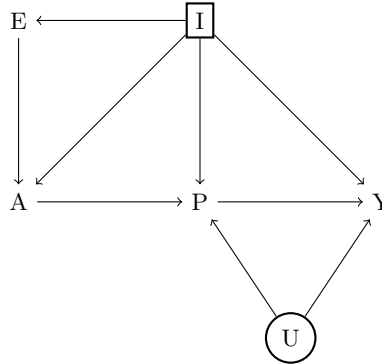
Question 1

In the causal diagram represented as Figure 1, denote A the eligibility for enrolling in a 401(k) plan, P actual participation in a 401(k) plan, Y accumulated net financial assets, E employment decisions, I income and U unobservable preference for savings. Participation in a 401(k) plan is affected by unobserved savings preferences U , eligibility A and possibly also income I . Estimating the unadjusted effect of P on Y therefore suffers from confounding.

However, given the information people base their employment decisions on income $I \rightarrow E$, but not directly on the availability of a 401(k) plan. Yet, the employment decision affects eligibility for a 401(k) plan as some employers do not offer such a plan $E \rightarrow A$. Income likely also has an effect on accumulated savings ($I \rightarrow Y$). Finally, we do not exclude the possibility that income also has direct effect on eligibility for a 401(k) plan ($I \rightarrow A$). Therefore, if we want to estimate the effect of 401(k) eligibility on accumulated savings, income is a confounder in the relation $A \rightarrow Y$ via the pathway $A \leftarrow I \rightarrow Y$, the pathway $A \leftarrow E \leftarrow I \rightarrow Y$ and the pathway $A \leftarrow I \rightarrow P \rightarrow Y$. After conditioning on I , all these pathways are blocked.

Furthermore, as people base their employment decisions on income I , but not on the presence of a 401(k) plan, unobservable savings preferences U are not related to the treatment A or employment decisions E . Despite the fact that savings preferences are related to accumulated savings, and actual participation status P , savings preferences are not a confounder of the effect of $A \rightarrow Y$. Therefore, we conclude that, conditional on income, we have blocked all confounding pathways, and thus $A \perp Y^a | I$.

Figure 1: Causal Diagram



Question 2



In this question, we use AIPW and **TMLE** to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets. We report both ATE and ATT because these effects could differ due to possible baseline differences and treatment heterogeneity. For example, consider factors that influence E (like household size) and assume that household size has treatment effect heterogeneity. The eligible and non-eligible groups could differ on this characteristic (household size), which leads to different estimates of the ATE and ATT.

Table 1 presents our results for the ATE estimation. We present five estimators. First, we present results for the AIPW estimator, using 5-fold sample splitting. This estimator relies on the estimation of the propensity score and an outcome regression model. We graphically depict the estimated propensity scores in Figure 2. There is considerable overlap between the treated and untreated groups. Still, as a robustness check, we present the AIPW when we exclude all individuals with EIC outliers. An outlying EIC is defined as below $Q_1 - 1.5 \times IQR$ or above $Q_3 + 1.5 \times IQR$. Third, we show the naive AIPW estimator without covariate adjustment. Fourth, we show the results for TMLE, again using 5-fold sample splitting. Fifth, we show the naive TMLE without covariate adjustment.

Our results for AIPW and TMLE align well, indicating ATEs of 7375 USD and 7540 USD, respectively. Eligibility for a 401(k) plan thus increases net financial assets with between 7375 and 7540 EUR. Both naive estimators show a much larger treatment effect, signalling the need for covariate adjustment. The result for the AIPW on the dataset without outliers is markedly smaller, which could be due to a more uniform treatment response after removing extreme observations. This suggests that the larger effects observed in the full sample may be driven by outliers, or it could indicate subgroup differences in treatment effects.

Table 1: ATE

Estimator	Estimate	Standard error	95% CI	p-value
AIPW	7375.18	1210.57	[5002.47; 9747.91]	0.000
AIPW (outliers removed)	4365.53	247.79	[3879.86; 4851.20]	0.000
AIPW (naive)	39118.69	1416.72	[36341.92; 41895.45]	0.000
TMLE	7540.93	1231.01	[5128.15; 9953.71]	0.000
TMLE (naive)	19559.34	1231.01	[17146.57; 21972.12]	0.000

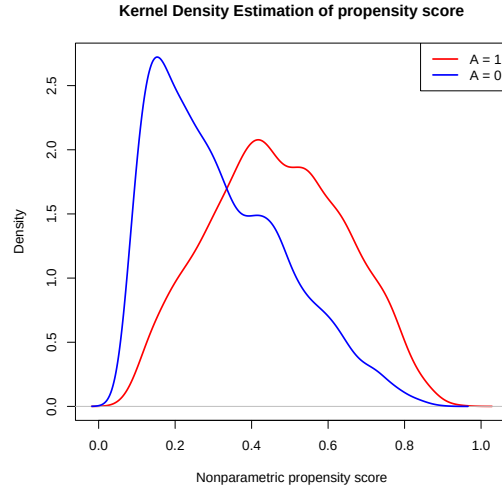


Figure 2: Propensity scores

Table 2 presents our results for the ATT estimations. Some estimators' results are larger than for the ATE, and some are smaller. This lack of clear distinction between the ATE and the ATT suggests that treatment effects may not differ significantly across eligible and non-eligible individuals, although the result is inconclusive. Remarkably, removing outliers of the influence curves of the AIPW ATT (of all treated individuals) does not influence the estimator, signalling that the outliers are located in the non-treated units.

Table 2: ATT

Estimator	Estimate	Standard error	95% CI	p-value
AIPW	10178.26	1478.45	[7280.49; 13076.03]	0.000
AIPW (outliers removed)	10768.20	777.28	[9244.71; 12291.68]	0.000
AIPW (naive)	19559.34	1412.85	[16790.16; 22328.53]	0.000
TMLE	5092.99	2012.51	[1668.674; 12338.79]	0.011
TMLE (naive)	7263.49	1413.96	[4492.13; 10034.86]	0.000

Question 3

In this question we want to test if $E\{\text{Cov}(A, Y|L)\} = 0$. To derive an estimator that does not suffer from plug-in bias, we set the sample average of the estimator's efficient influence curve equal to zero. The efficient influence curve of $\theta = E\{\text{Cov}(A, Y|L)\}$ is

$$\{A - E(A|L)\}\{Y - E(Y|L)\} - \theta \quad (1)$$

Setting the sample average for this equation equal to zero delivers the robust estimator:

$$\hat{\theta} = \sum_{i=1}^n \{A_i - \hat{E}(A_i|L_i)\}\{Y_i - \hat{E}(Y_i|L_i)\} \quad (2)$$

with $\hat{E}(A_i|L_i)$ an estimator of the propensity score $P(A = 1|L)$ and $\hat{E}(Y_i|L_i)$ an estimator for the outcome regression model $E(Y|L)$. All prediction models for the nuisance parameters are estimated using CV.SuperLearner, with 5-fold outer cross-validation. This guarantees the predictions are made on a set that is not used for training the algorithm. The results of our analysis are given in Table 3. We reject the null-hypothesis of no covariance between A and Y conditional on L at any reasonable significance level.

Table 3: Nonparametric Estimation Results

Estimator $\hat{\theta}$	Standard error	95% CI	p-value
1663.15	266.96	[1139.91; 2186.40]	0.000

Question 4

In this question, we use R-learner to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets in function of covariates. We divided the sample into five-folds and used four out of five as a training set for nuisance parameters (on all variables listed in question 2) and one as a test set to create honest causal predictions (age, income, and years of education). This algorithm was repeated until each fold had been once used as testing, i.e. 5-fold sample splitting.

Figure 3 shows the distribution of treatment effects obtained through the R-learner. Treatment effects vary between -7049 USD and 60,034 USD, with an average treatment effect of 7,077 USD, aligning close with our AIPW and TMLE estimators for the ATE. Figures 4 up to 6 show the marginal association between age, income and schooling on the one hand and the conditional average treatment effect, on the other hand. Our results indicate that the treatment effect of eligibility increases with age and income, while the effect of education is less clear.

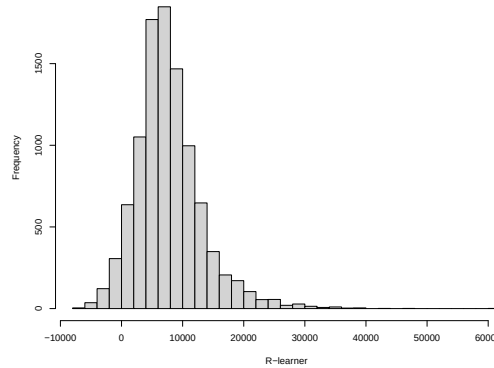


Figure 3: R-learner predictions

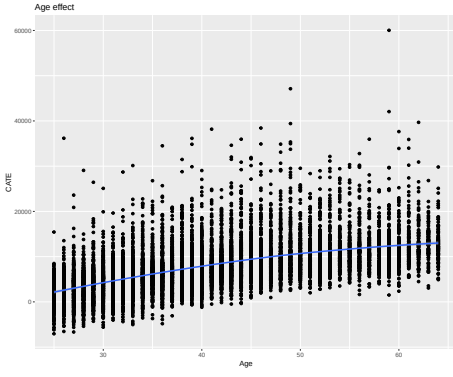


Figure 4: Age

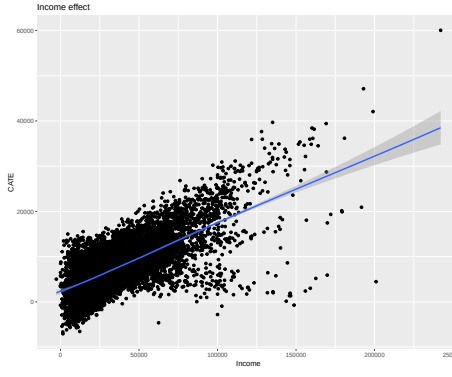


Figure 5: Income

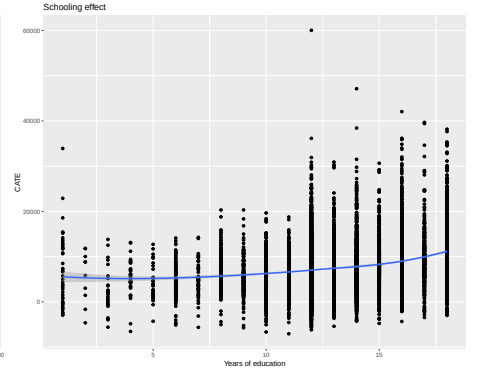


Figure 6: Schooling

Question 5

Part 1 In this question we predict the counterfactual outcome $\tilde{Y}_1$ using DR-learner. To this extent, we split the sample in a training and test set, each half of the sample. We compute the pseudo-outcome in the training set, for which we need predictions $\hat{Y}_i^1$ (the counterfactual outcome for treatment) and $\hat{p}_i$ (the propensity score). We estimate the propensity score uniquely on the training set using 5-fold cross-validation and honest predictions. The counterfactual outcome under treatment $\hat{Y}_i^1$ is also estimated using 5-fold cross validation and honest predictions. Both are implemented via the SuperLearner, with Generalized Linear Models and Random Forests as base learners. We then compute - still for the training set only - the pseudo-outcome C_i and regress these on age, income and years of education, again using SuperLearner and 5-fold cross-validation, now with a Generalized Linear Model. Using the obtained model, we predict the counterfactual outcomes $\tilde{Y}_i^1$ for all observations in the test set. The results are depicted in Figure 7, with counterfactual outcomes under treatment ranging between -37,788 USD and 225,239 USD.

Part 2 In the second part of this question, we estimate the counterfactual prediction error $E\{(Y^1 - \tilde{Y}^1)^2\}$ in the test set. The true value Y^1 is fundamentally unobserved. However, C_i is an unbiased estimator for Y^1 . To obtain the counterfactual prediction error, we thus substitute C_i in the expression for the counterfactual prediction error. This requires us to calculate C_i for the test set first, for which we use the same procedure as described above, estimating the propensity score and outcome regression using 5-fold cross validation and honest predictions, implemented via SuperLearner. We then compute $E\{(C_i - \tilde{Y}_i^1)^2\}$. The square root of this prediction error equals 80,952 USD, relatively large compared to the predictions. However, after removing observations with extreme predictions (pseudo-outcomes below -50,000 USD or above 250,000 USD), the square root prediction error reduces to 35,435 USD.

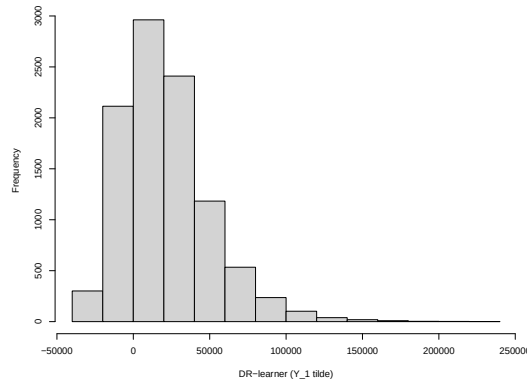


Figure 7: DR-learner